

Riley Brown

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EDUCATION

University of Nebraska-Lincoln

B.S. Mathematics and Computer Science, Lincoln, NE | Expected May 2023 - May 2027

- GPA 3.92/4.00
- University Honors Program

Budapest Semesters in Mathematics

Study abroad, mathematics, Budapest, Hungary | Jan 2026 - Jun 2026

EXPERIENCE

Research Assistant, pseudorandomness

North Carolina State University *advisor:* Professor Chin Ho Lee, Raleigh, NC | Present

Summer Researcher, algorithm design

Harvey Mudd College *advisor:* Professor Tim Randolph, Claremont, CA | Summer 2025

- Collaborated with a team of four students to analyze and improve exponential-time algorithms for the sumset recognition problem, an NP-complete problem in additive combinatorics
- Enhanced branching approaches and refined base case parameters for graph-theoretic reduction methods through systematic analysis of algorithmic complexity
- Presented findings at the REU symposium and continued independent exploration of the problem space

Investigating the Flow of Opinions in a Diagonal Grid Graph Using a Modified Deffuant-Weisbuch Model

University of Nebraska-Lincoln, Directed Reading Program *advisor:* Nick White, Lincoln, NE | Fall 2024

- Extended the project beyond its initial reading goals by implementing a modified Deffuant-Weisbuch model in Python
- Ran systematic parameter variation experiments showing that opinions invariably converge to consensus on diagonal grid graphs
- Proved the discovered properties and identified future directions on alternative graph topologies

Undergraduate Researcher, tiling theory

University of Nebraska-Lincoln *advisor:* Professor Tri Lai, Lincoln, NE | Summer 2024

- Applied Ciucu's Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
- Used Randall paths to determine hexagon-half parameters based on hole parity along the symmetric axis
- Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

Research Assistant, 3-uniform hypergraphic degree sequences, FPT algorithm feasibility

Budapest Semesters in Mathematics *advisor:* Professor Istvan Miklos, Budapest, Hungary | **[dates?]**

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln *advisor:* Professor Christine Kelley, Lincoln, NE | **[dates?]**

PROJECTS

Poker Trainer

A terminal and Tkinter drill trainer for poker skills: pot odds, combos, equity, bet sizing, push/fold and ICM, with an adaptive scheduler that weights drills by weakness and staleness. | python | 2026

SKILLS

Languages: Python, MATLAB, Java, SQL, LaTeX

Libraries and tools: NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow, Git, Jupyter

RELEVANT COURSEWORK

Mathematics: Bijective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Graph Theory, Probability Theory, Linear Algebra, **Algebra I**, Coding Theory

Computer Science: Development and Analysis of Algorithms, Automata, Computation, and Language, Data Structures and Algorithms, Data Analysis, Math for Machine Learning